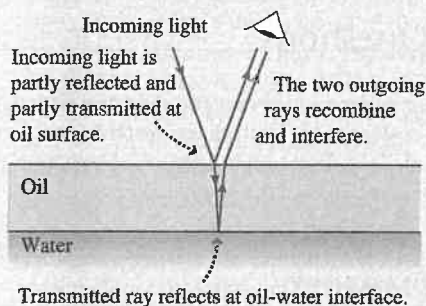


CHAPTER 22 SUMMARY

Interference

(Section 22.1) Interference of light waves can be **constructive interference**, which results in brighter light, or **destructive interference**, where the two waves cancel to produce a dark region. The Michelson interferometer uses interference of light to make exquisitely precise measurements of variations in distance and/or travel time for light. **Thin films** produce interference between light waves reflecting off their top and bottom surfaces. Reflection at an interface with a material of higher refractive index results in a 180° phase change.

Corrected wavelength with thin-film interference: $\lambda = \frac{\lambda_0}{n}$

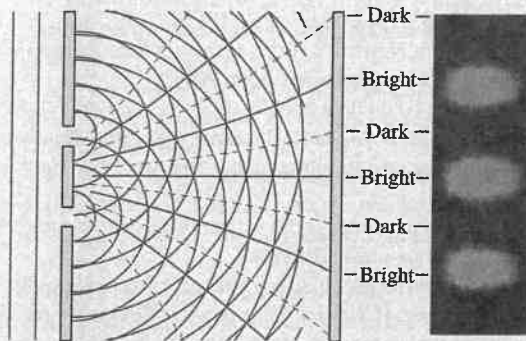


Double-Slit Interference

(Section 22.2) In **Young's double-slit experiment**, light passing through two slits interferes constructively and destructively, creating an interference pattern of alternating light and dark fringes on a distant screen.

Condition for bright fringes: $n\lambda = d \sin \theta$ ($n = 0, 1, 2, \dots$)

Distance of bright fringe from central maximum: $y = \frac{n\lambda L}{d}$



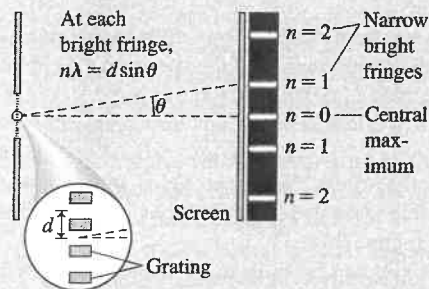
Diffraction

(Section 22.3) **Diffraction** is the bending of waves as they pass an obstacle or through an opening. Diffraction is more significant with obstacles or holes comparable in size to or smaller than the light's wavelength. A **diffraction grating** is a multiple-slit system that separates light precisely into its constituent wavelengths. Even **single-slit diffraction** produces a pattern of light and dark, as waves from different parts of the slit interfere. Regularly spaced atoms in a solid act as a diffraction grating for x rays.

Bright fringes from diffraction grating: $n\lambda = d \sin \theta$
($n = 0, 1, 2, \dots$)

Dark fringes from a single slit: $n\lambda = d \sin \theta$ ($n = 1, 2, \dots$)

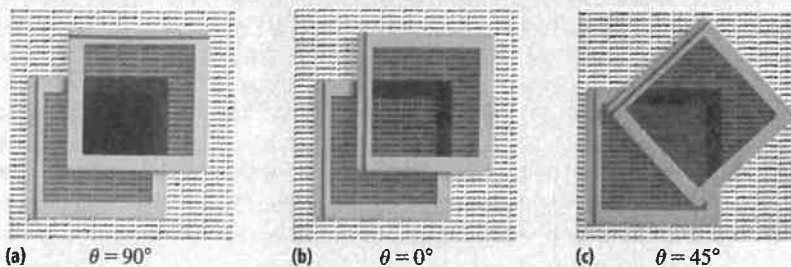
Rayleigh criterion: $\theta_{\min} = 1.22 \frac{\lambda}{D}$



Polarization and Scattering

(Section 22.4) Light is **polarized** when its electric field oscillates along a single axis. **Polarizers** pass light with a specific polarization; **Malus's law** describes the intensity of light passing through a polarizer as a function of its polarization direction. **Atmospheric scattering** of light is dominated by Rayleigh scattering, which favors scattering of shorter wavelengths.

Malus's law: $I = I_0 \cos^2 \theta$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

1. Why is interference easier to observe with sound than with light?
2. An antireflection coating needs its thickness matched to a particular wavelength of light. How might you make a coating that minimizes reflection throughout the visible spectrum, 400 nm to 700 nm?
3. How easy would it be to coat a military airplane's wings so they don't reflect 8-GHz radar waves?
4. Light with wavelength λ in a medium with refractive index n_1 strikes a thin film with index n_2 coating a material with index n_3 . For each of the following cases, what's the minimum film thickness that will maximize the reflected light? (a) $n_1 > n_2 > n_3$; (b) $n_1 > n_3 > n_2$; (c) $n_2 > n_3 > n_1$; (d) $n_2 > n_1 > n_3$.
5. The center spot in Newton's rings, formed where the two glass surfaces are in contact, is always dark. Why?
6. Explain how you could do a double-slit experiment with infrared or ultraviolet light. If you used the same apparatus, how would the interference pattern for ultraviolet differ from that made with infrared?
7. In a double-slit experiment, how is the fringe pattern affected if you triple the slit-to-screen distance?
8. A double-slit interference pattern stretches from one side of a screen to the other. If you increase the wavelength of the light used in the experiment, does the number of bright lines on the screen increase or decrease?
9. Why is it easier for AM radio signals to get around obstacles like buildings or hills?
10. You're observing a spectrum produced by a diffraction grating of spacing d . How will the appearance of the spectrum alter if you change to a grating of size (a) $0.6d$; (b) $1.2d$.
11. What will you see if you look at white light through a diffraction grating?
12. Your eyes can barely resolve two green lights some distance away. Will you be able to resolve two lights in the same locations if their colors are (a) blue; (b) red?
13. Describe two tests you can perform to tell whether your sunglasses are polarized.
14. Rank in increasing order the amount of atmospheric scattering experienced by these colors: yellow, green, orange, violet.

Multiple-Choice Problems

15. Red light ($\lambda = 630$ nm) is incident on an oil film ($n = 1.50$) on a puddle of water. What minimum oil thickness will result in no reflection? (a) 630 nm; (b) 420 nm; (c) 315 nm; (d) 210 nm.
16. An antireflection coating ($n = 1.42$) of thickness 82 nm is on a plastic eyeglass lens ($n = 1.68$). What visible wavelength will be least reflected? (a) 466 nm; (b) 510 nm; (c) 551 nm; (d) 617 nm.
17. A soap film ($n = 1.33$) with a thickness of 120 nm forms a bubble, with air on each side. What visible wavelength is most reflected from this bubble? (a) 420 nm; (b) 480 nm; (c) 560 nm; (d) 640 nm.
18. Light with wavelength 560 nm is sent through a pair of slits 1.0 mm apart. On a screen 1.8 m away, bright-line fringes are separated by (a) 0.10 mm; (b) 0.50 mm; (c) 1.0 mm; (d) 1.5 mm.
19. Red laser light ($\lambda = 632.8$ nm) passes through a pair of slits, producing an interference pattern on a screen 1.80 m away. If the distance from the central maximum to the third-order bright fringe on the screen is 1.10 cm, what's the slit spacing? (a) 0.20 mm; (b) 0.31 mm; (c) 0.42 mm; (d) 0.53 mm.
20. Light of unknown wavelength passes through a pair of slits 0.10 mm apart, producing interference on a screen 1.40 m away. If the distance between the central maximum and the first dark fringe of the interference pattern is 3.50 mm, what's the wavelength? (a) 650 nm; (b) 600 nm; (c) 550 nm; (d) 500 nm.
21. Monochromatic light with wavelength 480 nm passes through a diffraction grating with spacing $1.4 \mu\text{m}$. The angle at which a first-order bright line is observed is (a) 10° ; (b) 20° ; (c) 30° ; (d) 45° .
22. What's the smallest diffraction grating spacing that enables you to see the entire visible spectrum? (a) 400 nm; (b) 700 nm; (c) 1000 nm; (d) 1400 nm.
23. Two polarizing sheets have their transmission axes at 25° to each other. The portion of light from the first polarizer that makes it through the second is (a) 0.60; (b) 0.72; (c) 0.82; (d) 0.91.
24. A third polarizer is placed between a pair of crossed polarizers. Find the fraction of light coming from the first polarizer that makes it through the third if the angle between the transmission axes of the first two is 45° . (a) 0.50; (b) 0.25; (c) 0.12; (d) 0.

Problems

Section 22.1 Interference

For problems involving reflection, assume incoming light is normal to the surface unless indicated otherwise.

25. ■ Visible light is incident on an oil film ($n = 1.51$) coating a puddle of water. What minimum oil thickness will maximize the reflection of each of these colors: (a) blue (460 nm); (b) yellow (580 nm); (c) red (640 nm)?
26. ■ For what minimum oil thickness in the preceding problem will you see no reflection for the red light?
27. ■■ You pour some oil on water and, looking straight down, see a reflection of 440-nm violet light. What will you see if you (a) double and (b) triple the oil thickness?
28. BIO ■ **Reflection from eyeglasses.** An antireflection coating ($n = 1.38$) of thickness 105 nm coats a plastic eyeglass lens ($n = 1.64$). For what visible wavelength will reflection be best minimized?
29. ■■ The soap film comprising a bubble is 102 nm thick and has refractive index 1.33. What visible wavelength is best reflected from this bubble? Assume air both inside and outside the bubble and a viewing angle normal to the bubble surface.
30. ■■ You're observing Newton's rings with 520-nm light. What minimum air gap produces a bright ring?
31. BIO ■■ **Protecting your eyes.** When illuminated by white light, a pair of reflective sunglasses appears blue, peaking at 480 nm. How thick is the reflective coating if its refractive index is 1.36?
32. BIO ■ **Avoiding UV.** Plastic with $n = 1.42$ is to be coated on sunglasses so eye-damaging near ultraviolet light (UVA; $\lambda = 380$ nm) will be reflected. What thickness coating should you use?

33. ■■ A Michelson interferometer uses 589-nm yellow sodium light. How far must one mirror be moved for the interference pattern to shift by 10 fringes?
34. ■■■ The Michelson interferometer is often used to measure the refractive index of gases. A transparent cell, initially evacuated, is placed in one arm of the interferometer and illuminated with monochromatic light with $\lambda = 540$ nm. (a) Explain why the interference fringes shift when air is let into the cell. (b) If the cell is 6.0 cm long, by how many bright fringes does the pattern shift when the cell fills with air ($n = 1.00029$) at 20°C and $P = 1.0$ atm?

Section 22.2 Double-Slit Interference

35. ■■ Yellow light with wavelength 589 nm passes through a pair of slits separated by 0.56 mm. What is the separation between bright-line fringes on a screen (a) 2.0 m; (b) 3.0 m away?
36. ■■ Violet light with $\lambda = 440$ nm strikes a double slit and creates an interference pattern on a distant screen. At the position of the third-order bright fringe, how much farther is this fringe from the more distant slit than from the closer one? Does it matter how far away the screen is?
37. ■■ Green light with $\lambda = 540$ nm strikes a pair of slits separated by 0.085 mm. (a) Find the angular deviation of each of the first three fringes, relative to the central maximum. (b) Is the small-angle approximation justified?
38. ■■ Orange light with $\lambda = 615$ nm strikes a pair of slits separated by 0.580 mm. On a screen 1.20 m away, what's the distance between (a) the two second-order bright fringes; (b) the central maximum and one of the fourth dark fringes; (c) the third bright fringe on one side of the central maximum and the third dark fringe on the other?
39. ■■ Monochromatic light of unknown wavelength passes through two slits separated by 0.12 mm, producing an interference pattern on a screen 1.55 m away. If the distance between the central maximum and the first dark fringe is 3.40 mm, what's the wavelength of the light?
40. ■■ A double-slit system with slit separation 0.340 mm is used to determine the wavelength of light passed by a filter. The resulting bright fringes are 2.58 mm apart on a screen 1.50 m away. (a) What's the wavelength? (b) What color is the filter?
41. ■■ White light shines through a far-red filter and the resulting 680-nm light passes through a double slit and produces 17 bright fringes across a distant screen. If the red filter is replaced by a violet one ($\lambda = 440$ nm), how many bright fringes will be on the screen?
42. ■■■ You're doing a double-slit experiment with two different colors. You notice that the second-order bright fringe using 680-nm red light is in the same place as the third-order bright fringe of the other color. (a) What's the wavelength of that color? (b) What color is it?
43. ■■ You have a white light source and two different filters: $\lambda = 480$ nm and $\lambda = 630$ nm. The first-order bright fringe from a double slit is 9.0 mm from the central maximum when you use the shorter wavelength. What's the corresponding distance for the longer wavelength?
44. ■■ Young's double-slit experiment can be done with waves throughout the electromagnetic spectrum. Suppose microwaves ($f = 100$ GHz) are sent through a pair of slits 25 cm apart, cut into a metal sheet. If you have a microwave detector 10 m on the other side of the slits, how far do you have to move the detector to go from maximum to minimum microwave intensity?
45. ■■ You're attempting a double-slit experiment with 10-nm x rays, using an x-ray detector 0.80 m from a pair of slits. If the detector is capable of resolving maxima separated by 0.15 mm, how close together must the slits be?

Section 22.3 Diffraction

46. ■■ A diffraction grating has 1.0- μm spacing. Find the first-order diffraction angles for the following wavelengths: (a) violet, 440 nm; (b) green, 540 nm; (c) red, 640 nm.
47. ■■ A diffraction grating has 1.25- μm spacing. (a) What wavelength is diffracted through a 26.1° angle? (b) Is this wavelength visible in second order? If so, at what angle?
48. ■■ (a) Find the smallest grating spacing that lets you see the entire visible spectrum. (b) What spacing lets you see the entire visible spectrum through second order?
49. ■■ You have a diffraction grating with spacing 1.40- μm that passes infrared, visible, and ultraviolet. Find the first-order diffraction angles of the following wavelengths: (a) ultraviolet, 300 nm; (b) visible, 550 nm; (c) infrared, 900 nm.
50. ■■ You're using a red laser with $\lambda = 632.8$ nm to illuminate a diffraction grating. (a) What spacing will allow you to see diffraction through fourth order? (b) If you can barely resolve the fourth-order line, at what angles do the first three orders appear?
51. ■■ A diffraction grating has a 1.0- μm spacing. What portion of the visible spectrum can be seen in (a) first order; (b) second order; (c) third order?
52. ■■■ You're using a diffraction grating to view the 400- to 700-nm visible spectrum. Suppose you can see the spectrum through third order. (a) Show that the first- and second-order spectra never overlap, regardless of the grating spacing. (b) Show that the second- and third-order spectra always overlap.
53. ■■ The bright yellow sodium line in the sodium spectrum is actually a pair of closely spaced lines at 589.0 nm and 589.6 nm. You observe the sodium spectrum using a diffraction grating with a spacing of 1300 nm. Find the angular separation between the two sodium lines (a) in first order; (b) in second order.
54. Monochromatic light with wavelength 560 nm passes through a single slit 2.50 μm wide and 2.00 m from a screen. Find the distance between the first- and second-order dark fringes on the screen.
55. ■■ In a single-slit diffraction pattern, the fourth-order minimum is 1.5° from the center. What's the slit width if the wavelength is (a) 450 nm and (b) 650 nm?
56. ■■ A 633-nm red laser is aimed through a 0.125-mm-diameter circular pinhole. In the resulting diffraction pattern, what's the angle from the central maximum to the first dark ring?
57. ■■ Monochromatic light $\lambda = 520$ nm is aimed through a pinhole, producing a diffraction pattern on a wall 1.75 m away. If the radius of the second dark ring is 2.50 mm, what's the pinhole's diameter?
58. ■■ Monochromatic light is aimed through a 0.24-mm-diameter circular hole. The resulting diffraction pattern appears on a screen, and its second bright ring has diameter 2.10 cm. How should the diameter of the hole be changed to increase the second bright ring's diameter to 3.00 cm?
59. ■■■ The brightest star in the sky is Sirius, which is actually a pair of stars 8.6 light years from Earth, separated from each other by about 3.0×10^{12} m. What minimum telescope diameter is needed to resolve the two stars with 550-nm light?
60. ■■■ When Venus is close to Earth it appears in a crescent phase. You can see this using a good pair of binoculars. (a) Use the astronomical data in Appendix E to estimate Venus's angular width when it's closest to Earth. (b) To see the crescent clearly, you need a resolution about half this angular width. What's the minimum diameter for your binocular lenses, assuming 550-nm light?

61. ■■■ The Hubble Space Telescope has a 2.4-m-diameter mirror. (a) What's Hubble's minimum angular resolution, using 550-nm light? (b) Hubble is used in the search for planets orbiting nearby stars. If a star is 10 light years distant, what's the minimum distance between planet and star if the two are to be resolved? Compare your answer with the distance from Earth to Sun.
62. ■■ The three principal visible spectral lines from hydrogen have wavelengths 434 nm, 486 nm, and 656 nm. The principal line of sodium is at 589 nm. A sodium lamp used to calibrate a diffraction grating shows the first-order sodium line at 25.0° from the central maximum. Find the angular positions of the three first-order hydrogen lines in hydrogen.
63. ■■ What's the spacing in a diffraction grating that produces a 10° separation between the two shortest-wavelength hydrogen lines in the preceding problem?

Section 22.4 Polarization and Scattering

64. ■ Unpolarized light passes through two polarizers. Find the fraction of light from the polarizer that gets through the second when the angle between their transmission axes is (a) 30° ; (b) 45° ; (c) 75° ; (d) 90° .
65. ■ Two polarizing sheets have their transmission axes at 35° . What fraction of light from the first polarizer makes it through the second?
66. ■■ Polarized light passes through a polarizer, and only 10% gets through. Find the angle between the light's electric field and the polarizer's transmission axis.
67. ■■ A 10-mW unpolarized laser beam passes through two polarizers, whose transmission axes differ by 30° . What's the beam power (a) in the region between the two polarizers and (b) after the second polarizer?
68. ■■ A 5-mW laser beam has its electric field vertical. If it passes through a polarizer with its transmission axis at 60° to the vertical, what's the power of the transmitted beam?
69. ■■■ A polarizer is slipped between two crossed polarizers. Find the fraction of the light from the first polarizer that makes it through the last one if the angle between the transmission axes of the first two is (a) 30° ; (b) 60° ; (c) 89° .
70. ■■ Unpolarized light is passed through two polarizers. The light intensities emerging from the two are I_1 and I_2 , respectively. Graph I_2/I_1 as a function of θ , where θ is the angle between the two transmission axes and varies from 0 to 180° .
71. ■■■ Three polarizers in sequence have transmission axes at 0° , 45° , and 90° . (a) What fraction of light from the first polarizer gets through the third? (b) How does the situation change if the first two polarizers are interchanged?

General Problems

72. ■■ The air gap in part of a Newton's rings apparatus varies from zero to $2.00\ \mu\text{m}$. How many bright rings can be observed in this region? Use an average visible wavelength of 550 nm.
73. ■■ A spy satellite is orbiting 300 km above Earth's surface. What diameter mirror is necessary for it to resolve the 15-cm-high numbers on a license plate with 550-nm light? Assume diffraction limits the optical resolution.
74. B10 ■■ **Antireflection glasses.** You're applying a 110-nm-thick antireflection coating to eyeglasses, hoping to limit reflection in the middle of the visible spectrum ($\lambda = 550\ \text{nm}$). (a) What should be the coating's refractive index, assuming it's less than that of the lenses? (b) If instead you use a coating with $n = 1.42$, what's its minimum thickness?

75. ■■■ In a Newton's rings demonstration using 540-nm light, the distance from the center of the rings to the first bright ring is 0.30 mm. If the bottom piece of glass is flat, what's the curvature radius of the top piece?
76. ■■ The Hubble Space Telescope has a 2.4-m-diameter mirror; its successor, the James Webb Space Telescope, will have a 6.4-m mirror. Among the largest telescopes on the drawing board is the ground-based European Extra Large Telescope, whose mirror will measure 42 m across. Assuming diffraction is the limiting factor, could any or all of these telescopes resolve a star and its planet 75 light years from Earth, if the planet is the same distance from the star as Earth is from the Sun and the light used has 600-nm wavelength?
77. ■■■ The signal from a 103.9-MHz FM radio station reflects from two buildings 35 m apart, effectively producing two coherent sources of the same signal. You're driving at 60 km/h along a road parallel to the line connecting the two buildings and 400 m away. As you pass closest to the two sources, how often do you hear the signal fade?
78. B10 ■■ **Resolution of the eye.** You're viewing small objects from a distance of 2.0 m. What's the smallest object you can resolve under each of the following conditions, with light at the middle of the visible spectrum, $\lambda = 550\ \text{nm}$? (a) The room is brightly lit, and your pupil diameter is 4.2 mm. (b) The lighting is dim, and your pupil has expanded to 9.5-mm diameter.
79. ■■■ Unpolarized light passes through a succession of three polarizers. Their orientations with respect to the vertical are 0° , 65° , and 45° , respectively. Find the fraction of the original light intensity remaining after each of the three polarizers.
80. ■■ You have a summer job with an aerospace firm providing instrumentation for a space probe to study the outer planets. The probe will carry an ultraviolet spectrometer with a 2.0-cm-wide grating ruled at 102 lines/mm. The spectrometer must be able to resolve spectral features only 1 mm apart in twelfth order, when observing at a wavelength of 155 nm. Does it meet these specifications?
81. ■■ You're investigating an oil spill for the state environmental protection agency. You have a sample of the slick on glass, and in the thinnest section of the slick you observe constructive interference with 580-nm light. The refractive indices of the oil and glass are 1.38 and 1.52, respectively. What do you report for the oil slick's thickness?

Answers to Chapter Questions

Answer to Chapter-Opening Question

The wave nature of light prevents us from resolving objects smaller than about the wavelength of light, and therefore the bits of information coded optically on a DVD or Blu-ray disc can't be too close. Ordinary DVDs use 650-nm red laser light, with bit spacing of $0.74\ \mu\text{m}$ and corresponding information content of 4.7 gigabytes (GB). The invention of inexpensive blue lasers enabled Blu-ray technology, which uses 405-nm light and a $0.32\text{-}\mu\text{m}$ bit spacing. This and other improvements let Blu-ray discs store 25 GB of information, enough for high-definition movies.

Answers to GOT IT? Questions

Section 22.1 (a) $\lambda/2$ and (c) $3\lambda/2$

Section 22.2 (a) blue; (b) red; (c) yellow

Section 22.3 700 nm

Section 22.4 (b) $>$ (a) $>$ (d) $>$ (c)